

Esthetic Numbers, Folded Paths, and Algebraic Diagonals

Proofs of the listed A377000 conjectures
and a uniform boundary-correction theorem

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Abstract

Let $T(q, k)$ count the positive base- q integers with k digits for which adjacent digits differ by one. The retrieved OEIS entry A377000 lists conjectures relating this array to differential compositions, linear recurrences, and the sequence A182555. We give self-contained proofs of these statements, with the diagonal identity as the principal target. The key bijection identifies digit words with reflection orbits of k -step walks on a path with q vertices.

The same viewpoint yields a broader theorem. On every fixed-offset diagonal $q = k + d$, the generating function belongs to $\mathbb{Q}(z, \sqrt{1 - 4z^2})$ and has algebraic degree exactly two. All these generating functions have the same irrational part, so their pairwise differences are rational. Below the diagonal, the correction to a central-binomial formula is eventually a polynomial of an explicit degree; we give its coefficients, the complete finite exceptional set, and a sharp threshold. We also identify the exact minimal recurrence of every row using Chebyshev polynomials and prove the suggested A206603 identification.

A standard-library Python verifier checks the results with exact arithmetic, including 75,381 equalities or nonsingularity checks and an exhaustive enumeration of 131,070 sign words. An additional symbolic verifier checks the resolvent and reduced denominators. The proofs, not the experiments, establish the mathematical statements.

Status and attribution. This note proves the conjectures *as stated in the retrieved OEIS entry*; it does not certify worldwide priority. The classical spectral formula is due to De Koninck and Doyon, and the differential-operation model is due to Malešević. In particular, mere existence of row recurrences was already an immediate consequence of known results. The manuscript has not undergone independent peer review or proof-assistant verification.

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1 The problem, the source, and the result

1.1 An explicit OEIS-listed target

For integers $q \geq 2$ and $k \geq 1$, define

$$T(q, k) = \#\{(d_1, \dots, d_k) : 1 \leq d_1 \leq q-1, \quad 0 \leq d_i \leq q-1, \quad |d_{i+1} - d_i| = 1\}. \quad (1.1)$$

These are the q -esthetic numbers of length k . The initial digit is nonzero; subsequent zero digits are allowed. Thus $T(q, 1) = q-1$, and $T(2, k) = 1$ for all positive k .

The array is recorded in [A377000](#), introduced by Paolo Xausa, with three conjectures attributed to Chai Wah Wu on 21 October 2024 [1]. In the notation of this note, they assert:

- C1.** For even q , $T(q, k)$ counts meaningful differential compositions of order k on $\mathbb{R}^{q-1}$.
- C2.** For every fixed q , the sequence $T(q, k)$ satisfies a linear recurrence; several particular recurrences are displayed.
- C3.** $T(q, q-2) = \text{A182555}(q-2)$, on the domain $q \geq 3$ where the length is positive.

The same entry suggests an identification of another diagonal with [A206603](#). We make its indexing precise and prove it as well.

The main chosen problem is **C3**. It is a sharply stated equality between a finite combinatorial count and a sequence specified by an algebraic generating function. Proving only a large initial segment would not settle it; the goal is an identity of formal power series.

1.2 What the literature check does and does not establish

De Koninck and Doyon already express $T(q, k)$ as a finite sum of powers of path eigenvalues [3, Eq. (17)]. Consequently, C2's recurrence *existence* follows at once from prior work, even though the OEIS entry labels it a conjecture. We therefore do not present that part as a new discovery. Our recurrence treatment makes the minimal polynomial and the parity-dependent reduction explicit.

Malešević's paper defines meaningful differential compositions by matching input and output spaces and gives their adjacency rule [4]. We use that established definition and supply the bridge to esthetic digit words. The generating functions defining the two target sequences are recorded in their own entries [5, 6].

The sources and the revision history were consulted on 19 September 2026. The retrieved history lists a 5 November 2025 link update as its newest revision [2]; the retrieved entry still labels C1–C3 as conjectures. No separate resolution was located in the targeted search. This is evidence for the status of the *listed problem*, not a proof that an equivalent result has never appeared elsewhere. Historical priority of the more general diagonal formulas would require an additional specialist literature review.

1.3 The main extension in one statement

Here is the structural result that goes beyond the originally selected diagonal. All generating functions in this note are formal series at zero, and square roots have constant term one.

Theorem 1.1 (Fixed-offset diagonals). *For every integer d , put*

$$G_d(z) = \sum_{k \geq \max(1, 2-d)} T(k+d, k) z^k.$$

There is an explicitly computable rational function $R_d(z) \in \mathbb{Q}(z)$ such that

$$G_d(z) = R_d(z) - \frac{\sqrt{1-4z^2}}{2(1-2z)^2}. \quad (1.2)$$

Thus G_d has algebraic degree exactly two over $\mathbb{Q}(z)$, and $G_d - G_e \in \mathbb{Q}(z)$ for every pair of integers d, e .

When $d \geq 0$, one has the particularly simple formula

$$G_d(z) = \frac{d+1-2dz-\sqrt{1-4z^2}}{2(1-2z)^2} - \frac{d}{2}. \quad (1.3)$$

For $d = -c < 0$, the rational correction is given completely in Theorem 7.3. Its infinite tail comes from a polynomial in k of degree $\lfloor (c-1)/2 \rfloor$.

The remainder of the paper proves this theorem and the OEIS identifications. The path bijection comes first, then the recurrence and differential-composition consequences, followed by the diagonal analysis. Readers interested only in C3 may read Sections 2 and 5.

2 A reflection-orbit bijection

Let P_q be the path with vertices $0, 1, \dots, q-1$, and let A_q be its adjacency matrix. Each undirected edge represents the two possible oriented transitions between neighboring vertices. The vector $\mathbf{1}$ has q entries, all equal to one.

Theorem 2.1 (The extra-step identity). *For all $q \geq 2$ and $k \geq 1$,*

$$T(q, k) = \frac{1}{2} \mathbf{1}^\top A_q^k \mathbf{1}. \quad (2.1)$$

More precisely, esthetic words of length k are in bijection with reflection orbits of k -step walks on P_q .

Proof. A k -step walk is a sequence $(w_0, \dots, w_k)$ of vertices whose consecutive differences have absolute value one. Reflect every vertex by the map $i \mapsto q-1-i$. This reverses the sign of the first step, so, because $k \geq 1$, no walk is fixed by reflection. Each orbit contains exactly one walk whose first step is upward.

Take this representative and delete w_0 . Since $w_1 = w_0 + 1$, the first remaining vertex satisfies $w_1 \geq 1$. The sequence $(w_1, \dots, w_k)$ therefore satisfies (1.1). Conversely, prepend $d_1 - 1$ to any word $(d_1, \dots, d_k)$ counted by $T(q, k)$. The prepended vertex lies in $\{0, \dots, q-2\}$, and the first step is upward. These constructions are inverse.

Finally, $\mathbf{1}^\top A_q^k \mathbf{1}$ counts all k -step walks, with all starting and ending vertices allowed. Every reflection orbit has size two, proving the formula. $\square$

The exponent k in (2.1) is important. A length- k digit word normally describes $k-1$ steps, with a restricted starting vertex. The bijection replaces that asymmetry by *one additional step* and unrestricted endpoints. It is this change of viewpoint that makes the diagonal formulas simple.

For generating functions only, we will use the extension

$$\tilde{T}(q, 0) = q/2, \quad \tilde{T}(q, k) = T(q, k) \quad (k \geq 1). \quad (2.2)$$

This is half the number of zero-step walks. It is *not* a claimed count of empty digit strings, and it need not be an integer.

3 Differential compositions and the first conjecture

3.1 The precise meaning of a composition

The term “meaningful” refers to a formally composable word of first-order operations: the codomain of one operation must match the domain of the next. It does not mean that the resulting differential operator is nonzero, nor does it identify two words that happen to define the same operator. This is the convention in Malešević’s construction [4].

Let $q = 2m$ and $N = q - 1 = 2m - 1$. For $0 \leq i \leq N$, identify differential forms of degrees i and $N - i$ with a common field type $V_{\min(i, N-i)}$. There are m types $V_0, \dots, V_{m-1}$. Exterior differentiation, transported through these identifications, gives one operation from $V_{\min(i-1, N-i+1)}$ to $V_{\min(i, N-i)}$ for each $1 \leq i \leq N$.

Its type-transition graph has one directed edge in each direction between successive types, and one loop at the final type:

$$V_0 \rightleftharpoons V_1 \rightleftharpoons \dots \rightleftharpoons V_{m-1}, \quad V_{m-1} \longrightarrow V_{m-1}.$$

The final loop contributes *one* allowed transition. No convention that counts an undirected loop twice is being used.

For example, in dimension three the edges are $\text{grad} : V_0 \rightarrow V_1$, $\text{curl} : V_1 \rightarrow V_1$, and $\text{div} : V_1 \rightarrow V_0$. There are five meaningful second-order words, including curl grad and div curl even though both vanish.

Theorem 3.1 (The differential-composition identification). *For even $q \geq 4$ and every $k \geq 1$, the number of meaningful words of k first-order differential operations on $\mathbb{R}^{q-1}$ is $T(q, k)$. The same equality holds for $q = 2$ under the natural one-dimensional interpretation consisting of the single derivative.*

Proof. Write $q = 2m$. Fold the vertex set of P_{2m} by reflection, mapping i to $\min(i, 2m - 1 - i)$. This produces exactly the type-transition graph just described. Every quotient vertex has two lifts.

Choose a walk in the quotient and choose one of the two lifts of its starting vertex. Its lift is then forced step by step. At an ordinary edge there is a unique neighbor mapping to the specified next vertex; at the terminal loop there is a unique central edge crossing to the other lift. Consequently, every quotient walk has exactly two lifts, which are reflections of one another.

A word of k composable operations is the same as a k -step quotient walk: the successive edges specify the operations and the vertices specify their types. Such words are therefore in bijection with reflection orbits of k -step walks on P_{2m} . Apply Theorem 2.1.

For $m = 1$, the quotient is a single vertex with one loop. There is one word of every positive length, matching $T(2, k) = 1$. $\square$

This proves C1. Notice that no calculation with derivatives is needed once the meaning of a composable word has been fixed. The statement is about a finite type graph, not an analytic identity between operators.

4 Every row: spectrum and exact minimal recurrence

4.1 The surviving path eigenvalues

Define

$$\theta_j = \frac{\pi j}{q+1}, \quad \lambda_j = 2 \cos \theta_j, \quad 1 \leq j \leq q.$$

The vectors with components

$$v_j(i) = \sqrt{\frac{2}{q+1}} \sin((i+1)\theta_j), \quad 0 \leq i < q,$$

form an orthonormal eigenbasis of A_q . Indeed, the eigenvalue equation is the sine addition formula, the endpoint values vanish at positions -1 and q , and summing the finite geometric series gives orthogonality and the normalization.

The finite sine sum is

$$\sum_{i=1}^q \sin(i\theta_j) = \begin{cases} 0, & j \text{ even}, \\ \cot(\theta_j/2), & j \text{ odd}. \end{cases} \quad (4.1)$$

For completeness, start from $\sum_{i=1}^q \sin(i\theta) = \sin(q\theta/2) \sin((q+1)\theta/2)/\sin(\theta/2)$. At $\theta = \theta_j$, the second sine vanishes for even j . For odd j , the product in the numerator becomes $\cos(\theta_j/2)$, giving (4.1).

Applying the spectral decomposition to (2.1) now yields

$$T(q, k) = \frac{1}{q+1} \sum_{\substack{1 \leq j \leq q \\ j \text{ odd} \\ \lambda_j \neq 0}} \cot^2(\theta_j/2) \lambda_j^k, \quad k \geq 1. \quad (4.2)$$

This is the De Koninck–Doyon formula in an equivalent half-angle form [3, Eq. (17)]. We have derived it again to make the consequences and the exceptional zero eigenvalue transparent.

4.2 Chebyshev formulas and minimality

Let T_m and U_m denote the Chebyshev polynomials characterized by

$$T_m(\cos \theta) = \cos(m\theta), \quad U_m(\cos \theta) = \frac{\sin((m+1)\theta)}{\sin \theta}.$$

Their elementary recurrences imply that $2T_m(x/2)$ for $m \geq 1$ and $U_m(x/2)$ are monic integer polynomials.

Theorem 4.1 (Exact minimal row recurrence). *The monic minimal characteristic polynomial for the positive-length sequence $k \mapsto T(q, k)$ is*

$$C_{2m}(x) = U_m(x/2) - U_{m-1}(x/2), \quad (4.3)$$

$$C_{2m-1}(x) = \frac{2T_m(x/2)}{x^{m \bmod 2}}. \quad (4.4)$$

Its degree d_q is

$$d_q = \begin{cases} q/2, & q \text{ even}, \\ (q+1)/2, & q \equiv 3 \pmod{4}, \\ (q-1)/2, & q \equiv 1 \pmod{4}. \end{cases} \quad (4.5)$$

Writing $C_q(x) = x^{d_q} + c_1 x^{d_q-1} + \cdots + c_{d_q}$, the recurrence is

$$T(q, k + d_q) + c_1 T(q, k + d_q - 1) + \cdots + c_{d_q} T(q, k) = 0 \quad (k \geq 1). \quad (4.6)$$

No lower-order constant-coefficient recurrence holds, even eventually.

Proof. Formula (4.2) is a sum of powers of distinct, nonzero numbers λ_j . Each coefficient $w_j = (q + 1)^{-1} \cot^2(\theta_j/2)$ is strictly positive. Consequently, the product of $x - \lambda_j$ over the indicated odd indices is an annihilating polynomial.

To prove minimality, suppose a polynomial p gives a recurrence valid for all $k \geq k_0 \geq 1$. Then

$$\sum_j w_j p(\lambda_j) \lambda_j^k = 0 \quad (k \geq k_0).$$

Use as many successive values of k as there are surviving eigenvalues. The resulting matrix is a Vandermonde matrix multiplied by a nonsingular diagonal matrix with entries $\lambda_j^{k_0}$. It is invertible, so $w_j p(\lambda_j) = 0$ for every j . Thus p vanishes at every surviving eigenvalue. This proves minimality of the product.

It remains to identify the product. If $q = 2m$, then

$$U_m(\cos \theta) - U_{m-1}(\cos \theta) = \frac{\cos((2m+1)\theta/2)}{\cos(\theta/2)}.$$

Its m zeros in the range under consideration occur at $\theta = (2a-1)\pi/(2m+1)$, $1 \leq a \leq m$. These are exactly the odd-index path eigenvalues, and none is zero. Monicity proves (4.3).

If $q = 2m-1$, the odd-index angles are $(2a-1)\pi/(2m)$, $1 \leq a \leq m$, precisely the zeros of $T_m(\cos \theta)$. A zero eigenvalue is present exactly when m is odd. It contributes nothing to (4.2) for $k \geq 1$, so its factor x must be removed. This gives (4.4) and the three degree cases. $\square$

q	$C_q(x)$
2	$x - 1$
3	$x^2 - 2$
4	$x^2 - x - 1$
5	$x^2 - 3$
6	$x^3 - x^2 - 2x + 1$
7	$x^4 - 4x^2 + 2$
8	$x^4 - x^3 - 3x^2 + 2x + 1$
9	$x^4 - 5x^2 + 5$
10	$x^5 - x^4 - 4x^3 + 3x^2 + 3x - 1$
11	$x^6 - 6x^4 + 9x^2 - 2$
12	$x^6 - x^5 - 5x^4 + 4x^3 + 6x^2 - 3x - 1$

The displayed OEIS recurrences for $q = 6, \dots, 12$ are therefore correct. The result proves C2 while making explicit what was already implicit in the classical spectral formula.

Corollary 4.2 (Reduced denominators and a Hankel certificate). *The reduced denominator, normalized to have constant term one, of $\sum_{k \geq 1} T(q, k) z^k$ is $z^{d_q} C_q(1/z)$. Moreover, for every $s \geq 1$,*

$$\det(T(q, s+i+j))_{0 \leq i, j < d_q} = \prod_a (w_a \lambda_a^s) \prod_{a < b} (\lambda_b - \lambda_a)^2 \neq 0, \quad (4.7)$$

where the indices run through the surviving eigenvalues.

Proof. Summing $w_a \lambda_a^k z^k$ over $k \geq 1$ gives $w_a \lambda_a z / (1 - \lambda_a z)$. Its pole cannot be canceled by the terms with other eigenvalues. This proves the denominator claim. The Hankel matrix factors as $V \operatorname{diag}(w_a \lambda_a^s) V^T$, where $V_{ia} = \lambda_a^i$. Taking its determinant gives (4.7). $\square$

5 Walk ranges and the diagonal conjecture

5.1 Translating an unrestricted walk into a finite path

Let $\varepsilon = (\varepsilon_1, \dots, \varepsilon_k) \in \{-1, 1\}^k$. Set $s_0 = 0$, $s_i = \varepsilon_1 + \dots + \varepsilon_i$, and define

$$L(\varepsilon) = \min_{0 \leq i \leq k} s_i, \quad U(\varepsilon) = \max_{0 \leq i \leq k} s_i, \quad \mathcal{R}(\varepsilon) = U(\varepsilon) - L(\varepsilon).$$

Write $x_+ = \max(x, 0)$.

Proposition 5.1 (Range formula). *For every $q \geq 2$, $k \geq 1$,*

$$T(q, k) = \frac{1}{2} \sum_{\varepsilon \in \{-1, 1\}^k} (q - \mathcal{R}(\varepsilon))_+. \quad (5.1)$$

If

$$M_k = \sum_{\varepsilon \in \{-1, 1\}^k} U(\varepsilon), \quad M_0 = 0, \quad (5.2)$$

then

$$T(q, k) = q2^{k-1} - M_k \quad (q \geq k, q \geq 2). \quad (5.3)$$

Proof. A translation by an integer a fits the unrestricted walk into $\{0, \dots, q-1\}$ exactly when $-L(\varepsilon) \leq a \leq q-1-U(\varepsilon)$. The number of such integers is $(q - \mathcal{R}(\varepsilon))_+$. Summing counts all walks on P_q ; divide by two using Theorem 2.1.

Since the range of a k -step walk is at most k , the positive-part operation can be removed for $q \geq k$. Sign reversal interchanges U and $-L$, so $\sum_{\varepsilon} \mathcal{R}(\varepsilon) = 2M_k$. Substitution proves (5.3). $\square$

This explains why a varying base is especially tractable near and above $q = k$. The width constraint has no nonlinear truncation there: only a single unrestricted-walk statistic remains.

5.2 The total maximum of binary walks

Introduce the formal series

$$r = r(z) = \frac{1 - \sqrt{1 - 4z^2}}{2z} = z + z^3 + 2z^5 + 5z^7 + \dots. \quad (5.4)$$

It is the unique series with zero constant term satisfying

$$r = z(1 + r^2), \quad z = \frac{r}{1 + r^2}. \quad (5.5)$$

Lemma 5.2 (Maximum generating function). *One has*

$$\sum_{k \geq 0} M_k z^k = \frac{r}{(1 - 2z)(1 - r)}. \quad (5.6)$$

Define $h_k = M_k + 2^{k-1}$, including $2^{-1} = 1/2$ at $k = 0$. Then

$$h_k = \begin{cases} (k + \frac{1}{2}) \binom{k}{k/2}, & k \text{ even}, \\ (k + 1) \binom{k}{(k-1)/2}, & k \text{ odd}. \end{cases} \quad (5.7)$$

Proof. The generating function for walks that first hit level 1 at their last step is $r(z)$. One way to see this is to remove the final upward step and reverse the signs: the remainder is a nonnegative excursion of even length. Such excursions satisfy the first-return decomposition $C(t) = 1 + tC(t)^2$; hence $r(z) = zC(z^2)$ and $r = z(1 + r^2)$.

A walk that first reaches level $a \geq 1$ decomposes uniquely into a successive first-passage segments, each climbing one new level. Its generating function is r^a . After that first visit, an unrestricted suffix has generating function $1/(1 - 2z)$. Therefore the generating function for walks whose maximum is at least a is $r^a/(1 - 2z)$.

For a nonnegative integer U , the identity $U = \sum_{a \geq 1} \mathbf{1}_{\{U \geq a\}}$ holds. Sum over walks, then sum the geometric series in a . All sums are locally finite as formal series, and the result is (5.6).

Using (5.5) to simplify now gives

$$\sum_{k \geq 0} h_k z^k = \frac{\sqrt{1 - 4z^2}}{2(1 - 2z)^2} = \frac{(1 + 2z)^2}{2(1 - 4z^2)^{3/2}}. \quad (5.8)$$

The elementary expansion

$$(1 - 4t)^{-3/2} = \sum_{n \geq 0} (2n + 1) \binom{2n}{n} t^n$$

follows by differentiating $(1 - 4t)^{-1/2}$, or directly from the binomial series. Extracting odd powers in (5.8) gives $h_{2n+1} = 2(2n + 1) \binom{2n}{n}$. Extracting even powers gives

$$h_{2n} = \frac{1}{2}(2n + 1) \binom{2n}{n} + 2(2n - 1) \binom{2n - 2}{n - 1} = (2n + \frac{1}{2}) \binom{2n}{n}$$

for $n \geq 1$, while $h_0 = 1/2$. These are precisely (5.7). □

5.3 A whole family above the diagonal

For every integer d and $k \geq 0$, define the algebraic baseline

$$B_d(k) = (k + d + 1)2^{k-1} - h_k. \quad (5.9)$$

In particular, $B_d(0) = d/2$. For negative d or invalid small bases, $B_d(k)$ is only an auxiliary expression, not a digit count.

Proposition 5.3 (Baseline generating function). *For every integer d ,*

$$F_d(z) := \sum_{k \geq 0} B_d(k) z^k = \frac{d + 1 - 2dz - \sqrt{1 - 4z^2}}{2(1 - 2z)^2}. \quad (5.10)$$

For $d \geq 0$ and valid positive lengths, $T(k + d, k) = B_d(k)$.

Proof. The coefficient formula follows from (5.3) and $M_k = h_k - 2^{k-1}$. For the generating function, sum the linear-exponential term in (5.9) and subtract (5.8):

$$\sum_{k \geq 0} (k + d + 1)2^{k-1} z^k = \frac{z}{(1 - 2z)^2} + \frac{d + 1}{2(1 - 2z)}.$$

This simplifies to (5.10). □

Theorem 5.4 (The selected diagonal conjecture). *For every $k \geq 1$,*

$$T(k+2, k) = \text{A182555}(k). \quad (5.11)$$

Equivalently, C3 holds for all $q \geq 3$.

Proof. Set $d = 2$ in Proposition 5.3. Since $B_2(0) = 1$,

$$1 + \sum_{k \geq 1} T(k+2, k)z^k = \frac{3 - 4z - \sqrt{1 - 4z^2}}{2(1 - 2z)^2}. \quad (5.12)$$

The right-hand side is the defining generating function of [A182555](#) [5]. Equality of formal power series proves equality of every coefficient, not merely a finite initial segment. $\square$

An explicit positive formula is also available. Let $\text{Cat}(j) = \binom{2j}{j}/(j+1)$. Since $(1 - \sqrt{1 - 4z^2})/2 = \sum_{j \geq 1} \text{Cat}(j-1)z^{2j}$,

$$T(k+2, k) = 2^k + \sum_{j=1}^{\lfloor k/2 \rfloor} \text{Cat}(j-1)(k-2j+1)2^{k-2j}. \quad (5.13)$$

This gives a finite-sum expression independent of the original digit recursion.

5.4 The suggested [A206603](#) diagonal

Theorem 5.5 (An addition-triangle diagonal). *For $k \geq 2$,*

$$T(k, k) = \text{A206603}(k), \quad \sum_{k \geq 2} T(k, k)z^k = \frac{1 - \sqrt{1 - 4z^2}}{2(1 - 2z)^2}. \quad (5.14)$$

Moreover,

$$T(k+2, k) - T(k, k) = 2^k \quad (k \geq 2). \quad (5.15)$$

Proof. Set $d = 0$ in (5.10). Its constant and linear coefficients vanish, so it is already the generating function of the valid diagonal. It matches the generating function recorded for [A206603](#) [6]. Subtract the two baseline formulas with $d = 2$ and $d = 0$ to obtain (5.15). $\square$

There is also a direct check against the *combinatorial definition* of [A206603](#), without taking its generating function on trust. In an addition triangle with $n+1$ entries in the base, the apex weights those entries by $\binom{n}{0}, \dots, \binom{n}{n}$. Pairwise swaps show that the maximum is attained by assigning the increasing base values $j - n/2$ to the increasing binomial weights. The sorted weights are $\binom{n}{\lfloor j/2 \rfloor}$, $0 \leq j \leq n$.

For $n = 2m$, pair equal weights to obtain

$$\sum_{j=0}^{m-1} (4j+1-2m) \binom{2m}{j} + m \binom{2m}{m} = (2m+1)2^{2m-1} - (2m+\frac{1}{2}) \binom{2m}{m}.$$

For $n = 2m+1$, the corresponding paired sum is

$$\sum_{j=0}^m (4j-2m) \binom{2m+1}{j} = (2m+2)2^{2m} - 2(2m+1) \binom{2m}{m}.$$

These identities follow from binomial symmetry and $j \binom{n}{j} = n \binom{n-1}{j-1}$. The two results are exactly $B_0(n)$ from (5.7)–(5.9). Thus the maximum apex and the esthetic diagonal agree directly as well.

6 An exact finite-width resolvent and reflection formula

For $q < k$, the positive-part operation in (5.1) matters. We now compute its effect exactly. This provides the generalization needed for negative-offset diagonals.

6.1 Solving the endpoint-value recurrence

For vertices indexed by $1, \dots, q$, let

$$v_i(z) = \sum_{k \geq 0} (A_q^k \mathbf{1})_i z^k.$$

Then

$$v_i = 1 + z(v_{i-1} + v_{i+1}), \quad v_0 = v_{q+1} = 0. \quad (6.1)$$

The equation has a unique solution in formal power series: its constant terms are fixed, and each successive coefficient is determined by the previous coefficients.

Proposition 6.1 (Exact row resolvent). *With $r = r(z)$ as in (5.4),*

$$v_i(z) = \frac{1 - (r^i + r^{q+1-i})/(1 + r^{q+1})}{1 - 2z}. \quad (6.2)$$

Consequently,

$$\begin{aligned} \mathcal{T}_q(z) &:= \frac{q}{2} + \sum_{k \geq 1} T(q, k) z^k \\ &= \frac{q}{2(1-2z)} - \frac{r(1-r^q)}{(1-2z)(1-r)(1+r^{q+1})}. \end{aligned} \quad (6.3)$$

Proof. The right-hand side of (6.2) vanishes at the two boundary indices. Its constant particular solution is $1/(1-2z)$. The remaining terms satisfy the homogeneous recurrence because $r^i = z(r^{i-1} + r^{i+1})$, and the same is true of r^{q+1-i} . Thus it solves (6.1), and uniqueness proves the assertion. Sum over i and divide by two, using $\sum_{i=1}^q r^i = r(1-r^q)/(1-r)$ and Theorem 2.1. $\square$

Although (6.3) is expressed using an algebraic series r , the whole expression is rational in z for fixed q , as is already clear from the finite matrix resolvent. It should not be confused with the genuinely nonrational *diagonal* generating functions.

Define

$$K(u) = \frac{(1+u)(1+u^2)}{(1-u)^3}. \quad (6.4)$$

Using $1-2z = (1-r)^2/(1+r^2)$, rewrite the row resolvent as

$$\mathcal{T}_q(z) = \frac{q}{2(1-2z)} - \frac{r}{(1-2z)(1-r)} + K(r) \frac{r^{q+1}}{1+r^{q+1}}. \quad (6.5)$$

The first two terms have coefficient $q2^{k-1} - M_k$. The third term is therefore precisely the finite-width correction.

6.2 A coefficient-extraction identity

Set

$$\beta_0 = 1, \quad \beta_t = 4t \quad (t \geq 1), \quad (6.6)$$

and for $k \geq 0$ define

$$S(k, e) = \begin{cases} \sum_{\ell=0}^{\lfloor e/2 \rfloor} \beta_{e-2\ell} \binom{k}{\ell}, & e \geq 0, \\ 0, & e < 0. \end{cases} \quad (6.7)$$

As usual, $\binom{k}{\ell} = 0$ when $\ell > k \geq 0$.

Lemma 6.2 (The correction kernel). *For integers $a \geq 1$ and $k \geq 0$,*

$$[z^k] K(r(z)) r(z)^a = [u^{k-a}] \frac{(1+u)^2}{(1-u)^2} (1+u^2)^k = S(k, k-a). \quad (6.8)$$

Proof. The change of variables $z = u/(1+u^2)$ is valid for formal residues, because its linear coefficient is one. Its differential is $dz = (1-u^2)(1+u^2)^{-2} du$. Therefore

$$\begin{aligned} [z^k] K(r(z)) r(z)^a &= \text{Res}_{u=0} K(u) u^a \frac{(1+u^2)^{k+1}}{u^{k+1}} \frac{1-u^2}{(1+u^2)^2} du \\ &= \text{Res}_{u=0} u^{a-k-1} \frac{(1+u)^2}{(1-u)^2} (1+u^2)^k du. \end{aligned}$$

This proves the first equality. Finally,

$$\frac{(1+u)^2}{(1-u)^2} = 1 + 4 \sum_{t \geq 1} t u^t = \sum_{t \geq 0} \beta_t u^t.$$

Multiplication by the binomial expansion of $(1+u^2)^k$ proves the second. $\square$

Theorem 6.3 (Universal finite reflection sum). *For every $q \geq 2$ and $k \geq 1$,*

$$T(q, k) = q2^{k-1} - M_k + \sum_{j=1}^{\lfloor k/(q+1) \rfloor} (-1)^{j-1} S(k, k-j(q+1)). \quad (6.9)$$

Equivalently, replace $q2^{k-1} - M_k$ by $(q+1)2^{k-1} - h_k$.

Proof. Expand the last term of (6.5) as

$$K(r) \frac{r^{q+1}}{1+r^{q+1}} = \sum_{j \geq 1} (-1)^{j-1} K(r) r^{j(q+1)}.$$

This is a well-defined formal series since r has order one. For a coefficient of degree k , only $j(q+1) \leq k$ can contribute. Apply Lemma 6.2 with $a = j(q+1)$ and use Lemma 5.2 for the remaining terms. $\square$

The formula is an identity for all widths, not an asymptotic expansion. In particular, the apparently alternating correction is always equal to the nonnegative quantity

$$\frac{1}{2} \sum_{\varepsilon \in \{-1, 1\}^k} (\mathcal{R}(\varepsilon) - q)_+, \quad (6.10)$$

because $(q-R)_+ = q-R + (R-q)_+$. It compensates for the negative contributions that the untruncated baseline would otherwise assign to walks of excessive range.

7 Polynomial boundary corrections and all diagonal generating functions

7.1 The eventual correction polynomial

For $c \geq 1$, define a polynomial in x by

$$P_c(x) = \sum_{\ell=0}^{\lfloor (c-1)/2 \rfloor} \beta_{c-1-2\ell} \binom{x}{\ell}, \quad \binom{x}{\ell} = \frac{x(x-1) \cdots (x-\ell+1)}{\ell!}. \quad (7.1)$$

Thus $P_c(k) = S(k, c-1)$ for nonnegative integers k .

Theorem 7.1 (Polynomial correction with a sharp threshold). *For $c \geq 1$ and*

$$k \geq K_c := \max(c+2, 2c-1), \quad (7.2)$$

one has

$$T(k-c, k) = B_{-c}(k) + P_c(k) = (k-c+1)2^{k-1} - h_k + P_c(k). \quad (7.3)$$

The degree of P_c is $\lfloor (c-1)/2 \rfloor$. For $c \geq 4$, the threshold is sharp in the following strong sense:

$$T(c-2, 2c-2) = B_{-c}(2c-2) + P_c(2c-2) - 1. \quad (7.4)$$

Proof. Put $q = k - c$ in (6.9). The requirement $q \geq 2$ is $k \geq c + 2$. The $j = 1$ term has coefficient index $k - (q + 1) = c - 1$, so it is $P_c(k)$. All terms with $j \geq 2$ vanish if $2(q + 1) > k$, equivalently $k > 2c - 2$. This proves the formula on exactly the range (7.2).

Write $p = \lfloor (c-1)/2 \rfloor$. The coefficient of $\binom{x}{p}$ is $\beta_0 = 1$ for odd c , and $\beta_1 = 4$ for even c . It is nonzero; the degree is therefore p , with leading coefficient $1/p!$ or $4/p!$, respectively.

For $c \geq 4$ take $k = 2c - 2$, so $q = c - 2 \geq 2$. Then $k = 2(q + 1)$. Besides the $j = 1$ term there is exactly one further term, namely the $j = 2$ contribution $-S(k, 0) = -1$. No higher term can occur. This proves (7.4). $\square$

c	$P_c(k)$	First guaranteed valid $k = K_c$
1	1	3
2	4	4
3	$k + 8$	5
4	$4k + 12$	7
5	$\binom{k}{2} + 8k + 16$	9
6	$4\binom{k}{2} + 12k + 20$	11
7	$\binom{k}{3} + 8\binom{k}{2} + 16k + 24$	13
8	$4\binom{k}{3} + 12\binom{k}{2} + 20k + 28$	15

Example 7.2 (Why the finite exception cannot be ignored). For $c = 4$, the first admissible length is $k = 6$, with base $q = 2$. Here $P_4(6) = 36$, while $B_{-4}(6) = 3 \cdot 32 - (6 + \frac{1}{2})\binom{6}{3} = 96 - 130 = -34$. Thus $B_{-4}(6) + P_4(6) = 2$, whereas $T(2, 6) = 1$. The missing -1 is exactly the second reflection term in (7.4). At every $k \geq 7$, the polynomial formula is exact.

7.2 An explicit formula including every exceptional coefficient

For $c \geq 1$ set

$$P_c^*(z) = \sum_{\ell=0}^{\lfloor (c-1)/2 \rfloor} \beta_{c-1-2\ell} \frac{z^\ell}{(1-z)^{\ell+1}}. \quad (7.5)$$

It is the generating function of $P_c(k)$, because $\sum_{k \geq 0} \binom{k}{\ell} z^k = z^\ell / (1-z)^{\ell+1}$. Define the finite polynomial

$$E_c(z) = \sum_{k=c+2}^{2c-2} z^k \sum_{j=2}^{\lfloor k/(k-c+1) \rfloor} (-1)^{j-1} S(k, k-j(k-c+1)), \quad (7.6)$$

with any empty sum equal to zero. In particular, $E_c = 0$ for $c \leq 3$.

Theorem 7.3 (Every fixed-offset generating function). *With the valid-domain convention of Theorem 1.1,*

$$G_d(z) = F_d(z) - d/2 \quad (d \geq 0), \quad (7.7)$$

and, for every $c \geq 1$,

$$G_{-c}(z) = F_{-c}(z) + P_c^*(z) - \sum_{k=0}^{c+1} (B_{-c}(k) + P_c(k)) z^k + E_c(z). \quad (7.8)$$

Consequently, all conclusions of Theorem 1.1 hold.

Proof. For $d \geq 0$, the baseline counts every valid positive-length term. Remove its constant term $d/2$. The only further potentially invalid positive length is $k = 1, d = 0$, but $B_0(1) = 0$, so it requires no additional subtraction. This proves (7.7).

For $d = -c$, the valid lengths begin at $c + 2$. Start with the series $F_{-c} + P_c^*$, whose coefficient of z^k is $B_{-c}(k) + P_c(k)$. Subtract all coefficients below the valid range; this is exactly the finite sum in (7.8). At valid lengths $k \geq 2c - 1$, Theorem 7.1 says that no more changes are needed. At the finitely many remaining valid lengths, Theorem 6.3 says that the missing terms are exactly those with $j \geq 2$ recorded in E_c . Thus every coefficient is correct.

All terms of these formulas other than the square root in F_d are rational. The square-root term is independent of d , proving (1.2) and rationality of every difference. It also proves algebraicity of degree at most two. The degree is *exactly* two: if G_d were rational, then (1.2) would make $\sqrt{1-4z^2}$ rational. But a square of a rational function has even order at every zero or pole, while $1-4z^2$ has simple zeros at $z = \pm 1/2$. This is impossible. $\square$

These formulas give a constructive procedure for $R_d(z)$, not only an existence statement. Once R_d is written down, a polynomial equation for G_d is

$$(2(1-2z)^2(G_d(z) - R_d(z)))^2 = 1 - 4z^2. \quad (7.9)$$

Multiplying by the squared denominator of R_d makes it a polynomial identity with coefficients in $\mathbb{Q}[z]$.

For the first three negative offsets, the exceptional reflection polynomial E_c vanishes, but the low-index subtraction must still be performed. Direct evaluation gives

$$\begin{aligned} G_{-1}(z) &= F_{-1}(z) + \frac{1}{1-z} - \frac{1}{2}, \\ G_{-2}(z) &= F_{-2}(z) + \frac{4}{1-z} - 3 - 2z - z^2, \\ G_{-3}(z) &= F_{-3}(z) + \frac{z}{(1-z)^2} + \frac{8}{1-z} \\ &\quad - \frac{13}{2} - 6z - 5z^2 - 3z^3 - z^4. \end{aligned}$$

These examples also show why the artificial length-zero extension must be separated from the combinatorial series.

8 Consequences and computational use

8.1 Diagonal differences satisfy constant-coefficient recurrences

Define $Q_d(k) = 0$ for $d \geq 0$, and $Q_{-c}(k) = P_c(k)$ for $c \geq 1$. The preceding results give, for every fixed pair d, e and all sufficiently large k ,

$$T(k+d, k) - T(k+e, k) = (d-e)2^{k-1} + Q_d(k) - Q_e(k). \quad (8.1)$$

Thus the nonrational component disappears not just at the generating function level, but directly in the exact coefficient formulas.

If the polynomial on the right has degree at most r , the sequence of differences is annihilated eventually by

$$(\mathcal{E} - 2)(\mathcal{E} - 1)^{r+1}, \quad (\mathcal{E}a)(k) = a(k+1).$$

When the polynomial is zero, the factor $(\mathcal{E} - 2)$ alone suffices. For two distinct nonnegative offsets, (8.1) is exact at every length where both bases are valid, and reduces to $(d-e)2^{k-1}$.

8.2 A common diagonal asymptotic scale

The exact central-binomial formulas show that fixed offsets share the same leading boundary loss. Stirling's expansion applied separately to the two parities gives

$$h_k = 2^k \sqrt{\frac{2k}{\pi}} \left(1 + \frac{1}{4k} + O(k^{-2}) \right). \quad (8.2)$$

For even k , insert $\binom{k}{k/2} = 2^k \sqrt{2/(\pi k)} (1 - 1/(4k) + O(k^{-2}))$ into (5.7). For odd k , use $h_k = 2k \binom{k-1}{(k-1)/2}$ and expand in $1/k$; the coefficient $1/(4k)$ is the same.

Therefore, for every fixed integer d ,

$$T(k+d, k) = (k+d+1)2^{k-1} - 2^k \sqrt{\frac{2k}{\pi}} + O_d(2^k k^{-1/2}). \quad (8.3)$$

For negative offsets, the polynomial $Q_d(k)$ is exponentially smaller than the displayed error. When exact low-order effects matter, it should instead be retained explicitly through (7.3). The asymptotic estimate is a consequence of an exact formula, not the basis of the proof of a diagonal identity.

8.3 Which exact algorithm to use

The digit dynamic program starts with endpoint counts $(0, 1, \dots, 1)$ and repeatedly replaces the count at each vertex by the sum at its neighbors. It uses $O(qk)$ integer additions and $O(q)$ stored integers to produce a length- k count. The folded type graph halves the state count for even q .

For a fixed base and a very large isolated index, the minimal recurrence can be used with polynomial reduction modulo C_q . Binary powering computes $x^k \bmod C_q(x)$ in $O(d_q^2 \log k)$ arithmetic operations using ordinary polynomial multiplication, after which the result is a linear combination of initial terms. This is an arithmetic-operation bound, *not* a logarithmic bit-complexity claim: the output itself has a number of bits growing with k .

For fixed offsets, the central-binomial formula with the explicit polynomial correction is usually the most direct description. It requires no table over all smaller bases. Conversely, the universal reflection sum can involve large canceling quantities when the base is small; it is valuable as an exact formula and proof device, but is not asserted to be the fastest numerical method in every regime.

9 Exact verification and reproducible artifacts

9.1 Independent models and test coverage

The file `verify.py` uses only the Python standard library. The primary counts come from the endpoint digit recurrence, not from the new closed forms. Separate checks use unrestricted path walks, the folded graph, and Malešević's operation-index adjacency rule

$$M_{ij} = 1 \iff j = i + 1 \text{ or } i + j = N + 1, \quad 1 \leq i, j \leq N.$$

This last model counts words by $\mathbf{1}^T M^{k-1} \mathbf{1}$ and is independent of the type-vertex implementation.

The complete test run performed for this package passed the following checks. An “equality” refers to one exact comparison, not a floating-point tolerance test.

Check	Number	Range or interpretation
Reflection-orbit count	24,750	$2 \leq q \leq 100, 1 \leq k \leq 250$
Minimal-polynomial recurrence	22,225	Same rows; every fitting window
Differential-operation graph	2,000	Even $2 \leq q \leq 40, k \leq 100$
Folded-path graph	2,000	Same even bases and lengths
Total maximum by enumeration	16	All sign words, $1 \leq k \leq 16$
Range formula	320	$2 \leq q \leq 21, 1 \leq k \leq 16$
Universal reflection formula	5,850	$2 \leq q \leq 40, 1 \leq k \leq 150$
Nonnegative-offset formula	3,249	$0 \leq d \leq 12$, valid $k \leq 250$
Eventual polynomial formula	6,627	$1 \leq c \leq 30$, valid tail $k \leq 250$
Complete negative-diagonal series	7,530	Includes every invalid or exceptional low index
Sharp threshold	27	$4 \leq c \leq 30$
A182555 generating function	250	Through index 250
A206603 generating function	249	Indices 2 through 250
Addition-triangle maximum	249	Independently sorted binomial weights
Nonzero minimal-size Hankel determinant	39	Bases 2 through 40
Total	75,381	Exact equalities or nonsingularity checks

The exhaustive part examines $\sum_{k=1}^{16} 2^k = 131,070$ sign words. The Hankel determinants use fraction-free integer elimination. The recorded run used Python 3.13.5; the verifier completed in about 2.7 seconds in the working environment. Timing depends on hardware and is not a complexity claim.

9.2 Symbolic checks and their limits

The optional file `symbolic_verify.py`, run with SymPy 1.14.0, inverts $I - zA_q$ exactly for $2 \leq q \leq 14$. It checks the row-resolvent formula after the rational substitution $z = u/(1 + u^2)$, verifies the reduced denominator against the Chebyshev minimal polynomial, and checks three scalar rational identities used in the coefficient extraction. All of these checks passed.

The verifier intentionally separates empirical validation from proof. No finite collection of tests establishes a statement for all q, k ; that role belongs to the bijections, formal-series identities, and minimality argument above. Likewise, neither the code nor the proofs can certify historical novelty. No Lean or other proof-assistant formalization is included or claimed.

9.3 Contents of the archive

The archive contains the editable $\text{\LaTeX}$ source and compiled PDF, the two verifier programs, a build file, a source-provenance note, and machine-readable results. The generated data include:

- row values for $2 \leq q \leq 40$ and $1 \leq k \leq 100$;
- diagonal values for offsets $-12 \leq d \leq 12$ through length 200;
- all minimal polynomials for $2 \leq q \leq 100$ and the exact Hankel determinants used in the finite checks.

Run `python3 verify.py` from the archive directory to regenerate the standard-library results. The `README.md` supplies the complete build and reproduction commands. No external source papers or font files are redistributed.

10 Conclusion

The chosen OEIS-listed diagonal conjecture has a short conceptual resolution: append an initial upward step, pass to reflection orbits, and translate unrestricted binary walks into a finite path. At width $q \geq k$, the translation count is linear in the walk range, and a first-passage generating function evaluates the required total maximum. This proves C3 by exact generating-function equality.

The same construction proves C1 and gives explicit minimal recurrences for C2, with appropriate attribution to the known spectral theory. More importantly, it extends the diagonal identity to every fixed integer offset. Negative offsets introduce an explicit polynomial boundary correction after a sharp threshold, and only finitely many additional reflection terms before it. The resulting generating functions all share one quadratic irrational component.

The mathematical claims in this note are supported by complete arguments and reproducible exact checks. Their publication history is a separate question: the paper should be regarded as a proposed self-contained resolution and extension of the retrieved OEIS conjectures, not as a certificate of priority.

A Compact statements suitable for an OEIS update

The following formulas summarize the results without depending on the auxiliary derivations. They are a draft for review, not a claim that an OEIS edit has been submitted.

For $q \geq 2$ and $k \geq 1$, let A_q be the adjacency matrix of the path on q vertices. Then

$$T(q, k) = \frac{1}{2} \mathbf{1}^T A_q^k \mathbf{1}.$$

For even q , folding the path by reflection gives the type graph of meaningful differential compositions on $\mathbb{R}^{q-1}$, proving Conjecture 1.

The minimal row characteristic polynomials are

$$C_{2m}(x) = U_m(x/2) - U_{m-1}(x/2), \quad C_{2m-1}(x) = 2T_m(x/2)/x^{m \bmod 2}.$$

They prove the listed row recurrences and specify the exact minimal orders. Recurrence existence already follows from the known spectral formula.

For $q \geq k \geq 1$ and $q \geq 2$,

$$T(q, k) = (q+1)2^{k-1} - \begin{cases} (k + \frac{1}{2}) \binom{k}{k/2}, & k \text{ even,} \\ (k+1) \binom{k}{(k-1)/2}, & k \text{ odd.} \end{cases}$$

In particular,

$$1 + \sum_{k \geq 1} T(k+2, k)z^k = \frac{3 - 4z - \sqrt{1 - 4z^2}}{2(1 - 2z)^2},$$

proving $T(q, q-2) = \text{A182555}(q-2)$ for $q \geq 3$. Also $T(k, k) = \text{A206603}(k)$ for $k \geq 2$. All fixed-offset diagonals $q = k + d$, including negative d , have algebraic generating functions of degree two, with the explicit formulas in Theorem 7.3.

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